

ZHANGWU, China

WMO: 542360

Lat: 42.3842N

Lon: 122.5492E

Elev: 276

StdP: 14.55

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-2.8	-22.5	1.6	4.8	-18.8	2.0	5.6	25.2	26.3	21.8	23.2	3.9	320	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	89.5	71.5	87.0	71.5	85.0	70.9	78.0	84.1	76.6	82.2	75.3	80.6	10.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.3	138.7	81.9	74.9	132.1	80.1	73.6	126.3	79.0	41.9	84.3	40.5	82.4	39.1	80.7	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.8	21.7	18.5		-12.4	93.7	5.4	3.2	-16.3	96.0	-19.5	97.9	-22.5	99.6	-26.5	101.9	
				DB												
				WB	-13.0	80.4	5.3	1.6	-16.8	81.6	-19.9	82.6	-22.9	83.5	-26.7	84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	12.1	20.8	34.2	50.3	63.4	71.9	76.4	74.2	64.2	49.6	31.2	16.3
	DBStd	23.65	7.76	8.76	8.47	7.54	6.22	4.69	3.59	4.47	6.04	8.21	10.03	7.99
	HDD50	4296	1175	817	492	86	1	0	0	1	110	569	1045	
	HDD65	7473	1640	1237	954	442	104	5	0	2	83	481	1015	1510
	CDD50	3271	0	0	2	95	418	657	820	751	428	96	4	0
	CDD65	973	0	0	0	1	55	212	355	288	60	2	0	0
	CDH74	7833	0	0	0	28	605	1721	2816	2150	500	13	0	0
	CDH80	2239	0	0	0	1	141	573	855	591	77	1	0	0
Wind	WSAvg	8.1	6.7	8.0	9.6	11.0	10.1	8.4	7.5	6.5	6.8	7.7	7.7	6.6
Precipitation	PrecAvg	20.2	0.1	0.1	0.3	1.0	1.6	3.1	5.9	4.2	2.4	1.0	0.3	0.1
	PrecMax	29.3	0.4	0.6	2.4	4.5	4.1	6.5	13.1	10.0	8.5	3.7	1.0	0.8
	PrecMin	13.0	0.0	0.0	0.0	0.1	0.6	0.7	1.6	0.6	0.2	0.0	0.0	0.0
	PrecStd	4.2	0.1	0.2	0.5	0.8	1.0	1.5	2.9	2.3	1.7	0.9	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.6	48.7	63.6	77.7	88.3	93.1	92.6	90.7	85.4	75.7	61.0	43.0
		MCWB	30.5	36.2	47.0	55.2	62.4	67.1	74.4	74.7	68.4	61.2	51.5	38.8
	2%	DB	32.5	43.3	57.4	73.0	83.7	89.2	89.4	87.5	82.3	70.8	55.3	37.3
		MCWB	26.9	33.8	43.2	53.8	61.2	66.8	74.5	74.7	66.5	57.8	48.2	32.0
	5%	DB	28.9	39.1	52.7	68.6	80.6	86.1	86.9	85.4	79.4	67.2	51.1	33.3
		MCWB	24.0	31.6	40.2	51.3	59.1	66.6	73.9	73.7	64.7	55.9	44.2	29.0
	10%	DB	25.3	35.1	48.5	64.6	77.2	83.2	84.7	83.5	76.7	63.7	46.9	29.7
		MCWB	21.0	28.7	37.6	48.9	57.6	66.2	73.5	72.6	63.6	53.5	41.1	25.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.2	38.6	49.2	60.1	67.3	74.3	80.6	80.0	74.0	65.1	54.2	40.4
		MCDB	35.0	44.8	59.2	70.8	80.8	83.8	89.1	86.3	80.3	71.9	58.7	42.1
	2%	WB	27.4	35.1	44.7	56.4	64.7	72.0	78.4	78.1	70.8	61.7	49.4	33.4
		MCDB	31.4	41.7	54.7	68.1	77.5	81.3	84.9	83.9	77.8	67.2	54.0	36.1
	5%	WB	24.2	31.9	41.5	53.6	62.6	70.7	77.1	76.6	68.8	59.1	45.5	29.1
		MCDB	28.5	38.2	50.8	64.6	74.2	79.3	82.7	82.3	75.1	64.5	49.2	33.0
	10%	WB	21.0	29.1	38.5	51.0	60.5	69.3	75.9	75.3	66.8	56.1	41.2	25.2
		MCDB	25.4	35.0	47.2	62.0	72.1	77.3	81.3	80.5	72.8	61.4	46.6	29.4
Mean Daily Temperature Range	5% DB	MDBR	19.4	19.7	19.5	21.1	21.1	17.9	14.6	16.1	21.9	21.1	18.0	18.4
		MCDBR	23.3	23.7	26.2	27.6	27.5	24.6	18.3	18.3	24.7	24.9	22.6	21.8
		MCWBR	18.6	16.6	15.6	13.6	10.8	8.3	7.3	7.8	11.2	13.4	15.8	17.9
	5% WB	MCDBR	22.8	21.9	23.7	22.4	22.4	17.6	13.7	14.1	18.6	20.0	19.1	21.3
		MCWBR	18.8	16.3	15.0	12.4	10.6	7.5	7.2	7.3	10.5	13.2	15.6	18.4
Clear-Sky Solar Irradiance	taub	0.334	0.392	0.451	0.556	0.624	0.704	0.681	0.597	0.522	0.508	0.411	0.350	
	taud	2.103	1.967	1.875	1.693	1.613	1.541	1.634	1.786	1.879	1.817	1.978	2.072	
	Ebn at Noon	244	246	246	230	218	201	204	217	223	203	209	224	
	Edh at Noon	33	45	56	73	81	88	79	66	56	52	38	32	
All-Sky Solar Radiation	RadAvg	757	1051	1419	1694	1896	1735	1637	1593	1443	1070	745	631	
	RadStd	62	69	86	115	143	146	148	103	68	70	60	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	-1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page